

UBC Test Rocket Documentation

Fin Flutter, Separation

This doc covers the testing side of the UBC test rocket project. If you've already gone through the design phase from Kiara's Slides ([P How to make a rocket 👍 \(part 1\).pptx](#)

[P How to make a rocket 👍 \(part 2\).pptx](#)) — OpenRocket sims, CAD work, and picking a motor — this is the next step.

Now that we know what the rocket looks like and how it's powered, we need to make sure it actually holds up when it flies. We'll walk through three main areas: fin flutter calculations to make sure our fins don't shake apart at high speeds, separation testing to confirm everything detaches when it's supposed to.

Fin Flutter

What is it?

Fin flutter is when your fins start vibrating uncontrollably due to aerodynamic forces during flight. If the rocket hits a speed where this happens, the fins can shake themselves apart - which is obviously not great. So we need to figure out at what speed flutter actually kicks in and make sure we stay below it.

The flutter velocity changes depending on where the rocket is in its flight path. Altitude plays a big role here as you go higher, air pressure and the speed of sound both drop, which affects how the fins interact with the airflow.

FLUTTER: The Flutter velocity computed by AeroRocket's Microsoft Excel spreadsheet uses Equation-18, a flutter prediction derived in NACA TN 4197 on page 14. Equation-18 is a simplified flutter criterion for the most common type of flutter that is known as fin bending-torsion flutter. In this equation, G is the shear modulus of a solid-section fin, a is the speed of sound at the simulation altitude, P is the atmospheric pressure at the simulation altitude, P_0 is the sea level atmospheric pressure, AR is the aspect ratio of the fin equal to $\frac{b^2}{s}$, λ is the ratio of fin tip to fin root equal to $\frac{c_r}{c_t}$ and b is the semi-span of the fin. The flutter velocity for low aspect ratio fins that are solid and tapered toward the tip follows.

$$U_F = a \sqrt{\frac{G}{\frac{39.3 AR^3}{\left(\frac{t}{c_r}\right)^3 (AR + 2)} \left(\frac{\lambda + 1}{2}\right) \left(\frac{P}{P_0}\right)}}$$

$$P = P_b \cdot \left[1 + \frac{L_b}{T_b} \cdot (h - h_b) \right]^{\frac{-g_0 M}{R L_b}} \quad (2)$$

where,

P_b = static pressure (pressure at sea level) [Pa]

T_b = standard temperature (temperature at sea level)
[K]

L_b = standard temperature lapse rate [K/m] = -0.0065
[K/m]

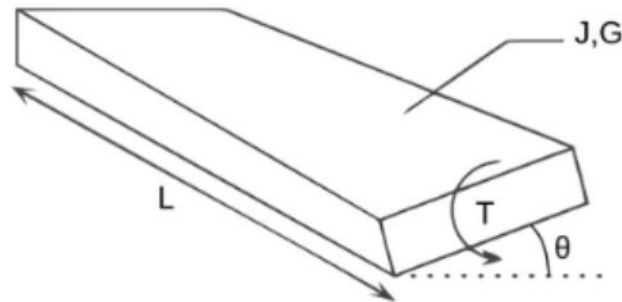
h = height about sea level [m]

h_b = height at the bottom of atmospheric layer [m]

R = universal gas constant = $8.31432 \left[\frac{N \cdot m}{mol \cdot K} \right]$

g_0 = gravitational acceleration constant = $9.80665 \left[\frac{m}{s^2} \right]$

M = molar mass of Earth's air = 0.0289644 [kg/mol]



Formula TLDR

At its core, it depends on the fin dimensions (aspect ratio, thickness, chord lengths) and how the atmosphere changes with altitude - specifically air pressure and the speed of sound. As well as the shear modulus which can be measured experimentally.

Using the fin dimensions from our CAD model and OpenRocket and plugging them into the spreadsheet along with the speed of sound and pressure at each altitude, we get a flutter velocity for every point in the flight.

Trapezoidal Fin Set configuration

Component name:

General | Fin tabs | Override | Appearance | Comment

Number of fi...

Fin cant: °

Root chord: in

Tip cho... in

Heig... in

Sweep leng... in

Sweep ang... °

Fin cross sectio...

Thickne... in

Placement

Position relative ...

plus in

Fin rotation: °

Material

Component materi...

Component finish:

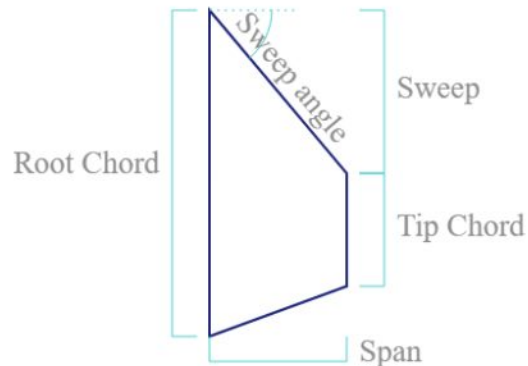
Root Fillets

Fillet radi... in

Fillet materi...

Component mass: 329 g

A single fin:



When using the sheet, input all the known dimensions from the open rocket. Input it on the sheet.

Also, experimentally measure the shear modulus (If they are 3D printed and you have as close to a material as possible). Measure the deflection of the most outward tip and the force you applied for it to be deflected. The one on the sheet is garbage data from fins that I assumed we had similar material properties to.

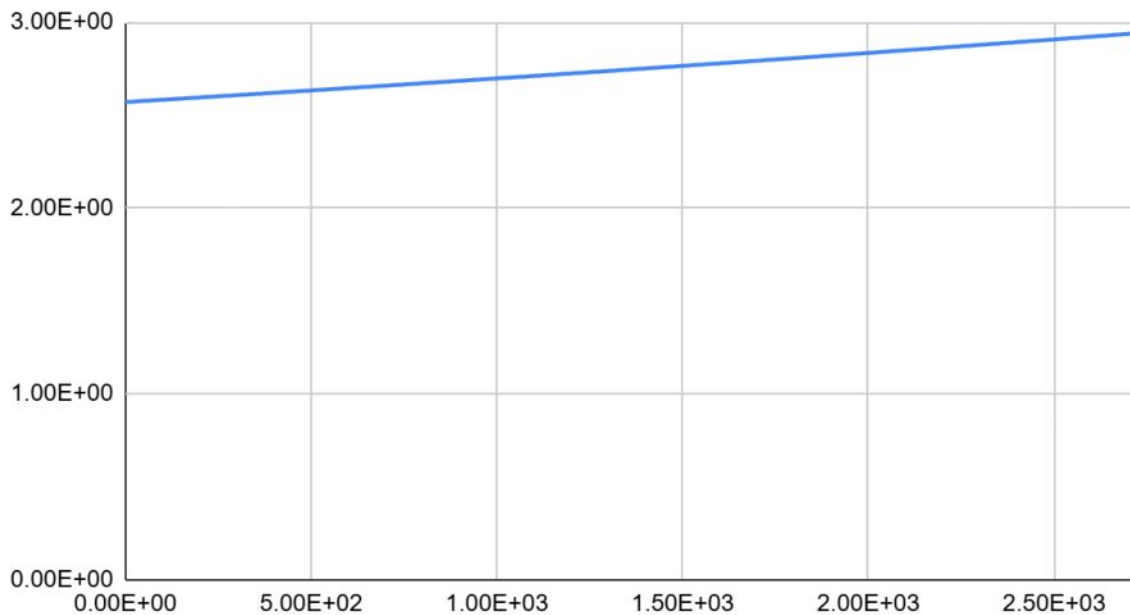
Or, if you are using a purely 3d printed fin, you can find values online (e.g. from Bambu).

Then from your open rocket, get simulation data on altitude, speed of sound at altitude and mach number (vehicle speed).

Afterwards, you will get graphs.

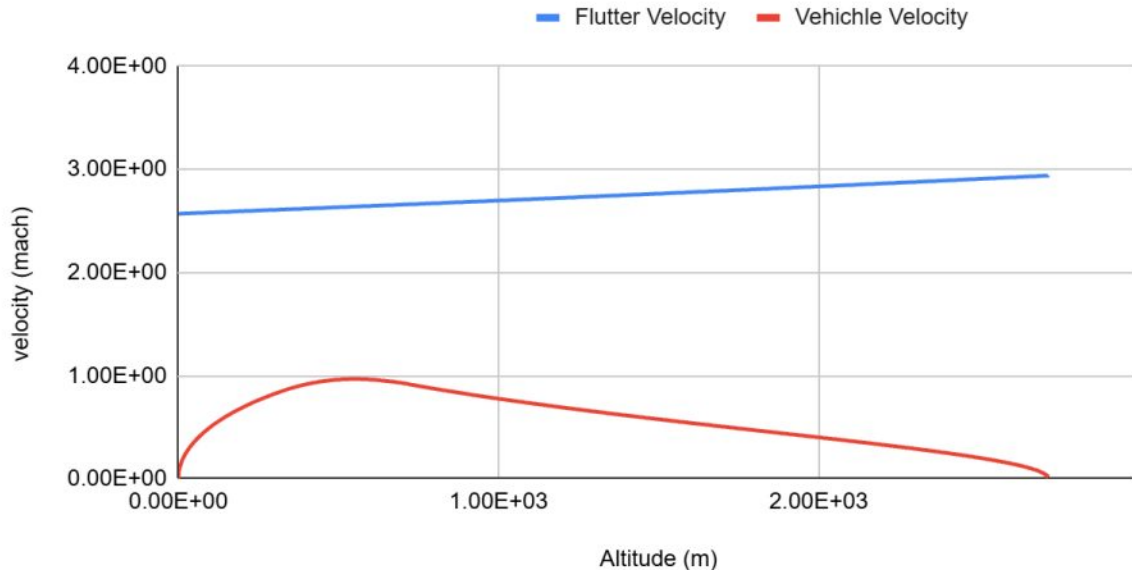
Flutter Speed vs Altitude

Flutter Velocity (Mach #) vs. Altitude(m)



Flutter Speed vs Vehicle Speed at Diff altitudes

Flutter Velocity vs Vehicle Velocity



Essentially, you want to ensure your graph of vehicle velocity never touches the flutter velocity. Otherwise, you are cooked 🤖.

Separation Test

When the rocket reaches apogee, it needs to split open so the parachute can deploy. This is done using a black powder charge (BP) that fires at the separation point between the body tubes. Too little powder and it won't open — too much and you risk damaging the rocket.

The goal of this testing is to find the right amount of black powder to get a clean separation every time. This section will cover:

- **Hardware** — the Egg Timer Quantum/Easy Mini and how it triggers the separation
- **Packing** — how the black powder, Ematches, and Centrifuge tubes are assembled together
- **Finding the optimal charge** — using a search approach to narrow down the minimum amount of BP needed for a clean separation

Hardware

Egg Timer Quantum Flight Computer	Easy Mini baro-only altimeter
 A green printed circuit board (PCB) for the Egg Timer Quantum Flight Computer. It features a black circular barometric pressure sensor in the center, a gold-colored WiFi module on the left, and various electronic components like resistors and capacitors. There are several pins and connectors along the edges.	 A green PCB for the Easy Mini baro-only altimeter. It has a large black circular barometric pressure sensor on the right side. A green terminal block with four screws is on the left. The board is labeled with 'Main Battery', 'Easy Mini', and 'Apogee Switch'. There are also some handwritten markings like '9018'.
Documentation: https://eggtimerrocketry.com/wp-content/uploads/2019/08/Eggtimer-Quantum-1_07A-1.pdf Can do Wifi Controlled detonation of charges.	Documentation: https://altusmetrum.org/AltOS/doc/easymini.html#_easymini Needs to be set at a certain pressure via usb?? (To be verified, last one broke)

Please do continuity checks with multi meters before connecting any batteries on the ports.

EggTimer Quantum

- The board you will use is called Eggtimer Quantum, these boards are wifi enabled which means you actually arm them (turn on the pyro channels, aka e-matches) using your phone which is what makes it useful for separation testing, since you control when the charges go off with your phone from a safe distance when in test mode
- **Batteries:** Works with a 2S/7.4V LiPo battery, 300 mAH or more recommended (Check with multimeter).
- **Connections:** To connect to the Eggtimer, first connect it to the battery recommended above for power, **it will beep 3 times which means everything is good**. Then you go into wifi on your phone and enter the corresponding password for that board. Example shown below:

Experiment

Finding the Optimal BP Charge

The goal is to find the minimum amount of black powder that gives a clean, forceful separation every time — not so little that it barely pops, but not so much that pieces fly 100m apart.

Start at **1g** and **2g** simultaneously — fire one test at each. This gives you a range right away:

- If **both are too weak**, jump up to **3g** and **4g**
- If **1g is too weak** and **2g works or is too strong**, you know the answer is somewhere between 1-2g. Now test **1.5g**
- Keep halving the interval between your bounds until you land on the optimal charge

Once you get a consistent "just right" result, fire it **2-3 more times** at that charge to confirm it's reliable.

You should have your answer in **3-4 tests** max.